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Beyond the Numbers: Predicting Olympic Medal's Future with Word2Vec and Ensemble Learning

Summary

In recent years, advancements in machine learning and statistical analysis have enabled researchers to leverage historical data to predict Olympic medal outcomes, providing a more scientifically grounded approach to forecasting Olympic results.

This paper employs the **Word2Vec method**, utilizing CBWM and Skip-gram algorithms, to convert complex text data into numerical vectors. We then build a **random forest regression model** consisting of 100 decision trees, which serves to predict the medal outcomes of countries in the 2028 Summer Olympics.

The model predicts that **93 countries** will win medals in 2028. Of the total medal standings for 2028, **28 countries** are predicted to **rise**, **42** to **fall**, with the remaining countries maintaining relatively stable rankings. Among these, the United States is expected to lead with 120 total medals, including 42 golds. More details are shown in main text.

Additionally, through probability mapping, we forecast that four countries have a 40% or greater chance of achieving a breakthrough in medal counts ranging from 0 to 1: Honduras-74.63%, Samoa-61.19%, Guinea-44.78%, Mali-43.28%.

We also conduct an error analysis of our predictions for the last five Summer Olympics, evaluating the prediction errors of each year using confusion matrices, root mean square error (RMSE), and mean square error (MSE). The RMSE errors (including for gold, silver, bronze, and total medals) and the corresponding confusion matrix percentage values are presented in the results.The model's **five-year average Accuracy** was calculated to be **89.21%**, its **Precision 89.56%**, its **Recall 84.62%**, and its $\mathbf{F_1}$ **score 86.98%**, based on five years of data. These findings indicate that the model demonstrates a high degree of proficiency in accurately predicting medal counts in the majority of cases.

Furthermore, we analyze the impact of a country's sports strengths, the benefits of being a host nation, and the influence of renowned coaches on medal outcomes. This provides actionable insights for future investments in the sports sectors of certain countries.

Our model evaluation indicates that it can accurately predict Olympic medal results, demonstrating strong robustness. Based on these predictions, we offer tailored recommendations to various national Olympic Committees, aiming to guide their sports programs towards the ideals of "Faster, Higher, Stronger."

Keywords: Word2vec, RF Regression, Medal Prediction, Host Effect, Great Coach Effect

Contents

1 Introduction

1.1 Background

At the 2024 Summer Olympics in Paris, spectators will will pay great attention to the "medal table" for each participating nation. Throughout the competition, the medal rankings of different countries are subject to fluctuations based on specific indicators. Consequently, the present study aims to develop a model to predict medal rankings, which possesses significant commercial potential.

1.2 Restatement of the Problem

- The attached data must be preprocessed and analyzed to identify the factors that will impact the medal prediction table, where the uncertainty and precision must be tested. The construction of a model is imperative to predict the outcomes of the medal table for the 2028 Summer Olympics in the United States of America. This model will elucidate the progression and regression of each nation's rankings in each event, as compared to the 2024 and previous editions.

- Concurrently, the observation that certain nations have advanced from the initial zero-medal status to a position where they have now won at least one medal is a salient phenomenon that merits our attention. Consequently, the development of a predictive model to ascertain the potential medal-winning nations and the probability of their success is imperative.

- It is imperative that our model take into account the conditions previously enumerated, including the number and variety of Olympic events, which are to be determined by the host nation. A rigorous examination of the relationship between events and the number of medals won by a country is imperative. Furthermore, the significance of individual sports in different countries and the impact of the events chosen by the host country on the outcome of the games must be thoroughly investigated.

- The impact of a "great coach" effect on the outcome of a game is significant. It is imperative to refine the predictive model by incorporating this effect into our model. A select group of countries should be identified, and projects that should be invested in "great coaching" should be determined. The results of this investment should then be evaluated.

- A subsequent analysis of other influential factors is to be conducted through the utilization of the model results. The development strategy corresponding to these factors must then be provided to the NOC.

1.3 Our Work

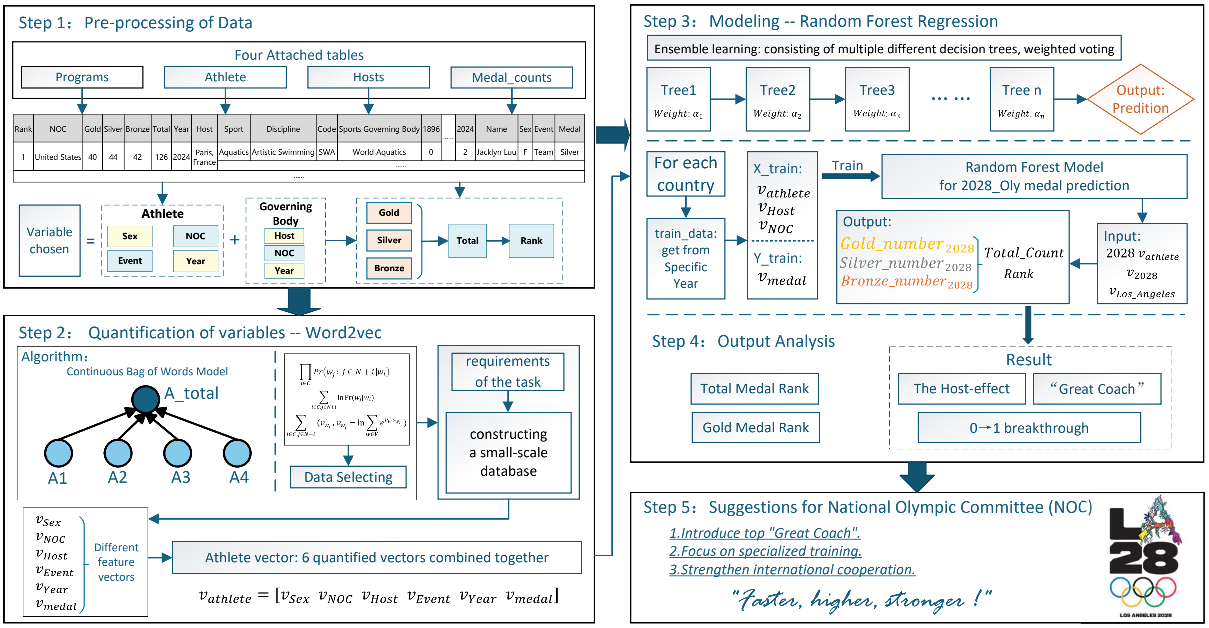
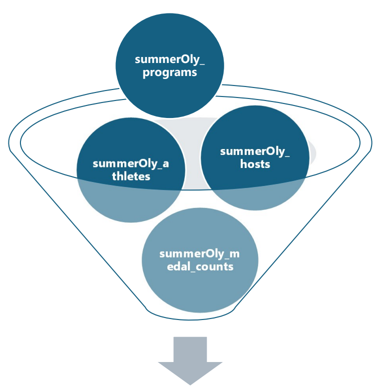


Figure 1: The overview of our work

2 Preparation for Modeling

First, it is possible to arrange the four tables in the Annex according to the configuration depicted in Figure 2 which provides a clearer elucidation of the relationships among them.



Rank	NOC	Gold	Silver	Bronze	Total	Year	Host	Sport	Discipline	Code	Sports Governing Body	1896	......	2024	Name	Sex	Event	Medal
1	United States	40	44	42	126	2024	Paris, France	Aquatics	Artistic Swimming	SWA	World Aquatics	0		2	Jacklyn Luu	F	Team	Silver
								......										
......																		

Figure 2: Relations between the attached tables

Then, we sorted out the relationship between Variables by using feature engineering and EDA after data cleaning.

2.1 Data Cleaning

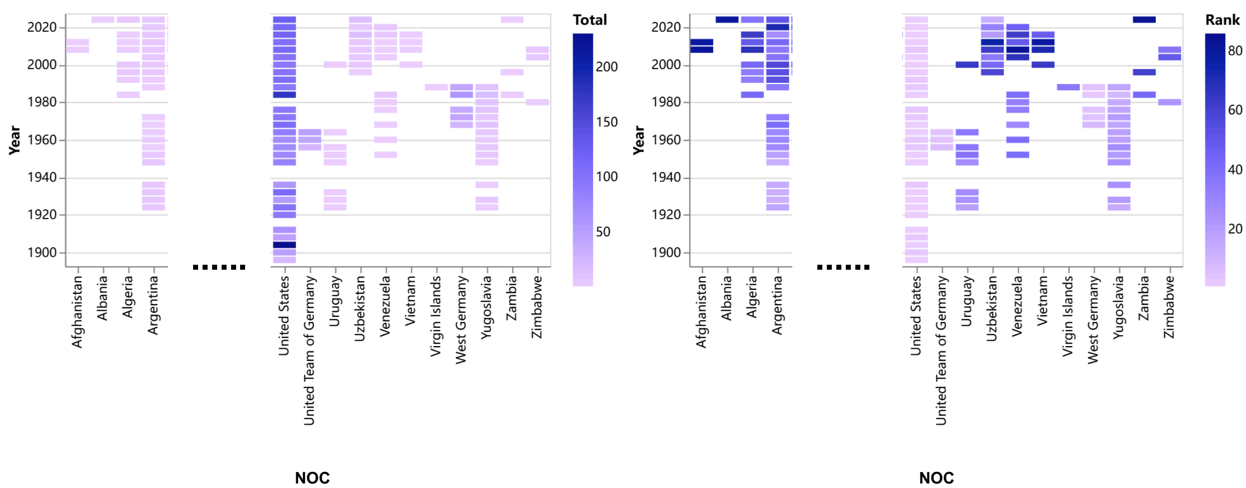
For the data in the attached table, we made a short process at first: we removed Skating and Ice Hockey from the "summerOly_program" which only belong to Winter Olympics these years.

Additionally, there are inconsistencies in the name of countries in "summerOly_medal_counts" and "summerOly_athlete". For example, both Roma and Rome have been standardized as Rome.

2.2 Feature Engineering & EDA

Next, to identify the factors which impact number of medals and find the relationship between them, we visualized the attached tables provided in compressed package.

Firstly, We draw Figure 13 based on "summerOly_medal_counts" which reflects the distribution of medal counts within the two-dimensional framework of time and nation. In Figure 13(a) the darker the color, the more medals the country has in that year; In Figure 13(b), the lighter the color, the more advanced the country is in that year. Among the figure for example, the United States always ranked first and have the most medals.



(a) (b)

Figure 3: Data pre-processing 1: **a)**Relationship between country, length of participation in the Olympic Games and total of medals won; **b)**Relationship between country, length of participation in the Olympic Games and rankings

Then we draw Figure 4(a) to build inner connections between medals, gold, silver, bronze and time series.From the figure, we discover that the total number of medals increases as the year increases. And the number of gold, silver and bronze increases as the year increases too. This also implies a potential possibility that as the time (Year) increases, the number of events increases. Figure 4(b) and (c) then explains a certain principle between Rank, Total and Gold, Silver, Bronze. That is, the higher the Rank, the higher the Totaland Gold which is not absolutely right in the reflection of the relationship between Rank and Silver or between Rank and Bronze. Therefore, we can conclude from Figure 6 that in each country Gold, Silver, and Bronze affect Total medals and Rank reflects the level of Total medals.

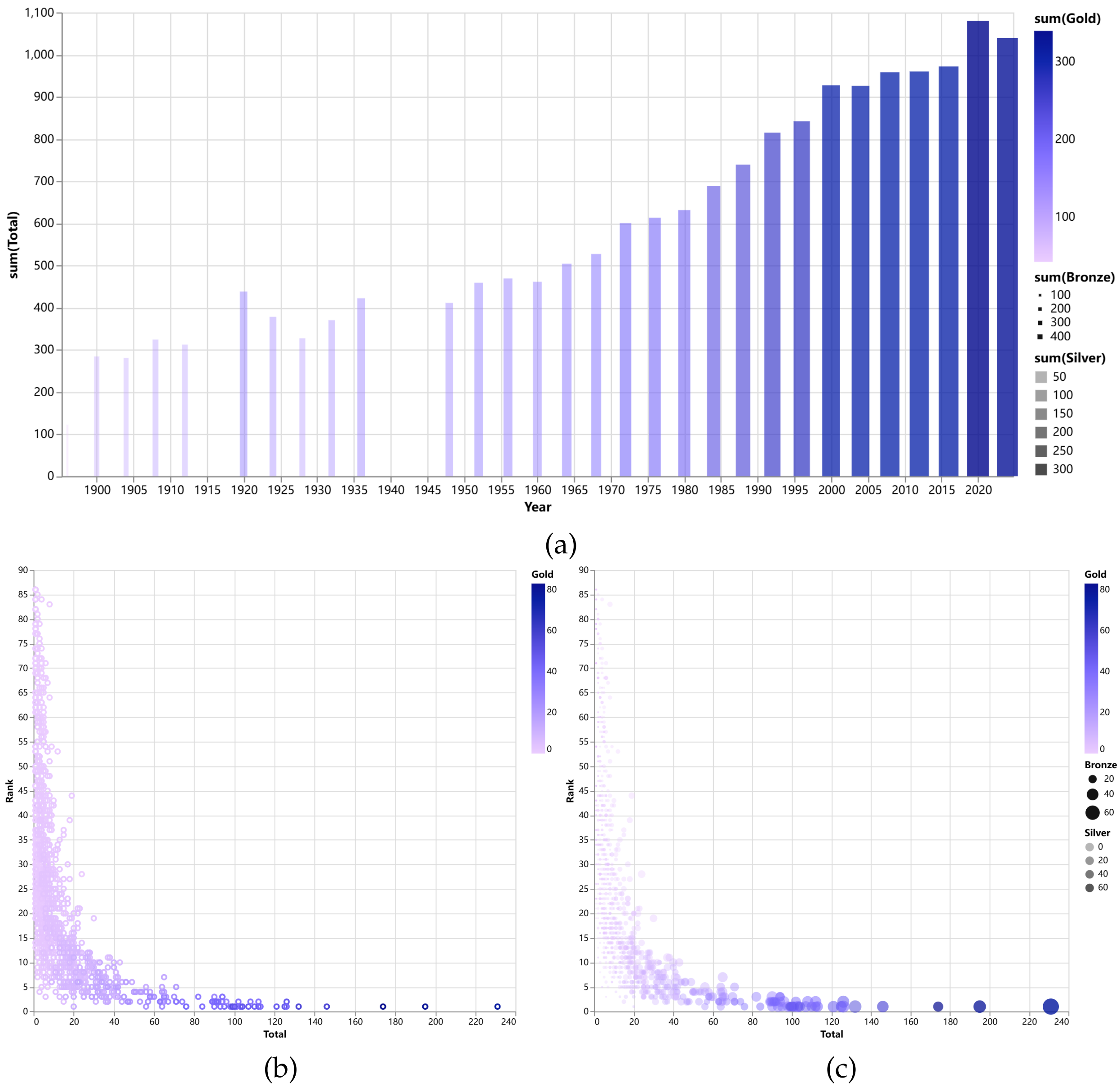


Figure 4: Data pre-processing 2: **a)**Relationship between medals, years and gold, silver and bronze medals; **b)**Relationship between medals, rankings and gold medals; **c)**Relationship between medals, rankings and gold, silver and bronze medals

After that, we analyze the table of "summerOly_athletes" which gives us list of information about each athelete who has attended Olympic before 2025. these information inluence atheletes' likelihood of obtaining Olympic medals.

And In the table of"summerOly_hosts", it is evident that a nation's status as a host nation exerts a substantial influence on its position within the medal table rankings, and this influence is reflected on atheletes' level directly. As illustrated in Heat Map 5, the diagonal line of the matrix indicates that when a country is hosting the Olympics, it will have a higher ranking. The figure displays four horizontal dark bars, which reflect the leading position of the United States.

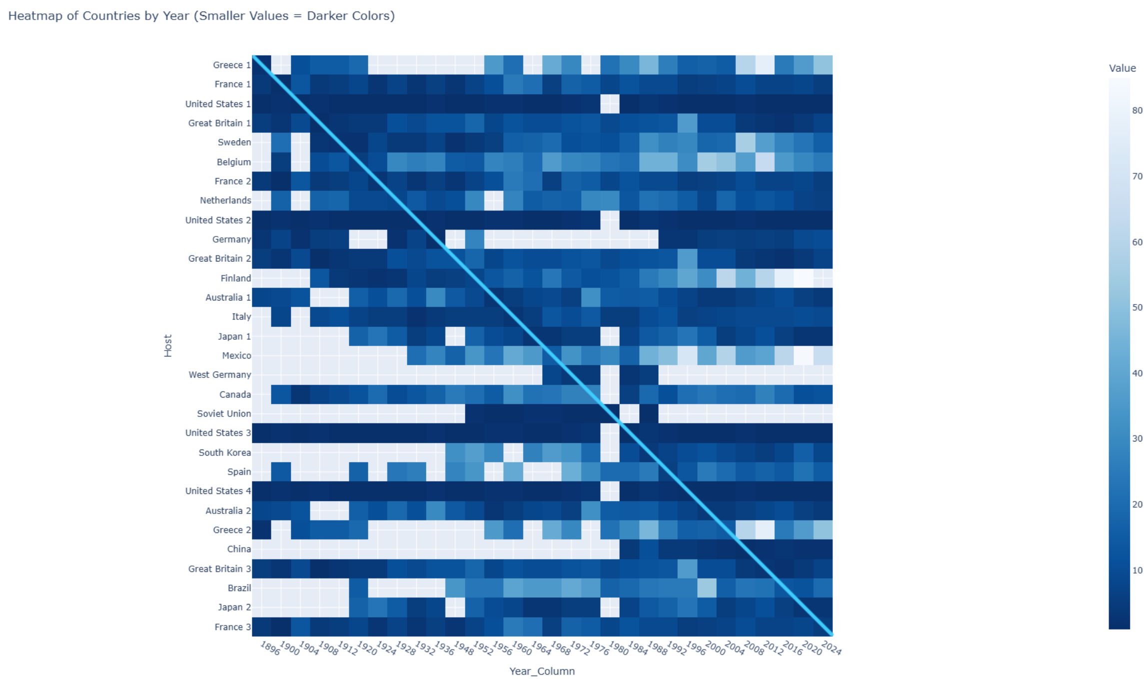


Figure 5: Host effect

Throughout all these analyses, we infer that the number of medal, Gold, Silver and Bronze are affected by atheletes, time, contries and host. which gives us a clearer control of variables' relationships.

2.3 Assumptions and Justification

To simplify the model, we make the following assumptions:

- We consider TEAM and NOC to be equivalent. That is, some athletes participate on an individual basis, which is still attributable to the State.
- In the Olympic medal count prediction model, we assume that the historical data of medal for each country has Inheritance, i.e., past data of medal are significant when predicting the number of medal in the future.
- when predicting each countries' number of medals, it is assumed that a nation's probability of attaining an Olympic medal is demonstrably associated with its economic development level, the advancement of sports infrastructure, the efficacy of

the athlete training system, and a myriad of additional factors. The combined effect of these factors is reflected in the variables of the provided data, enabling the prediction of outcomes through the utilization of the available data.

2.4 Notation

Table 1: Notations

Symbol	Description	Representation Space
$\boldsymbol{v}_{Host}$	Quantification of Host, see Figure9	$\mathbb{R}^{100}$
$\boldsymbol{v}_{NOC}$	Quantification of NOC, see Figure9	$\mathbb{R}^{100}$
$\boldsymbol{v}_{Event}$	Quantification of Event, see Figure9	$\mathbb{R}^{100}$
$\boldsymbol{v}_{Year}$	Quantification of Year, e.g.$[2, 0, 2, 4]$	$\mathbb{R}^{4}$
$\boldsymbol{v}_{medal}$	Quantification of Medal, e.g.$(0), (1), (2)$	$\mathbb{R}^{1}$
$\boldsymbol{v}_{Sex}$	Quantification of Sex, $[1, 0], [0, 1]$	$\mathbb{R}^{2}$
$\boldsymbol{v}_{athlete}$	$\boldsymbol{v}_{athlete} = (\boldsymbol{v}_{Sex}, \boldsymbol{v}_{NOC}, \boldsymbol{v}_{Host}, \boldsymbol{v}_{Event}, \boldsymbol{v}_{Year}, \boldsymbol{v}_{medal})$	$\mathbb{R}^{307}$
T_i	The i_{th} tree ($i \in \{1, 2, 3, ..., 100\}$)	1
α_i	The weight of the i_{th} tree	1

3 Modeling

3.1 Relationship Construction

By using the conclusion from Featuring Engineering and EDA, we can construct the relationship between all this variables: The Sport Governing Body, Sport, Discipline, and Event exhibit a hierarchical relationship, with the NOC directly influencing the Athlete and also impacting the Athlete through the Host. The Athlete directly influences the Event and also affects the Event through their Sex. The Year influences medal prediction through the Event, while the Year, Event, Athlete, and NOC all have a direct impact on medal prediction. The flowchart below shows these Relationships clearly.

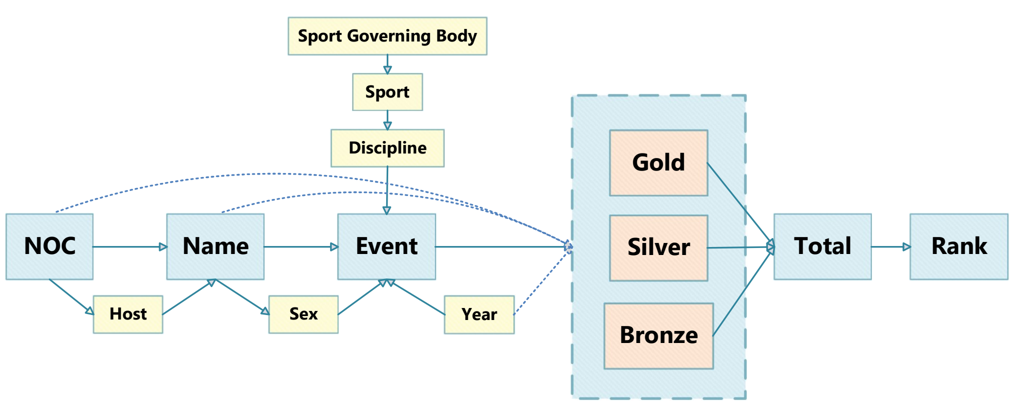


Figure 6: Variable analysis

3.2 Quantification of variables

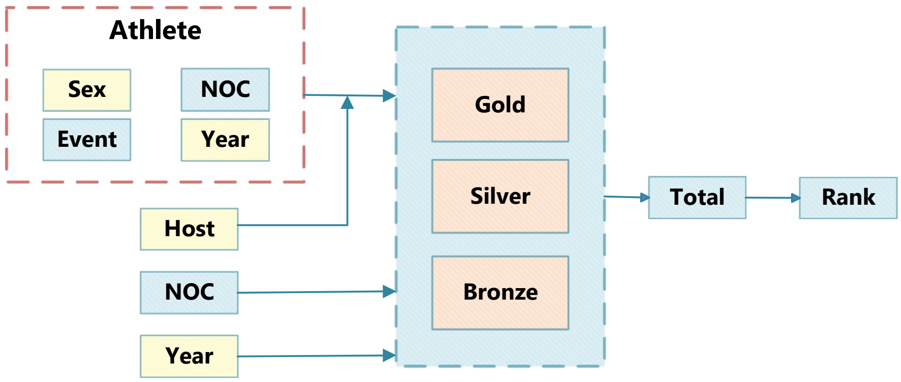


Figure 7: Quantitative variables

In the specific model, according to conclusions getting grom feature egineering and EDA above, by using Word2Vec which is introduced below, we can rearrange the relationship of those variables:

1. combine Sex, NOC, Event, Year together as the property of Athelete
2. combine Host and Athelete together as one part of influence on Medal prediction
3. remove the inner connections between NOC, Athelete, Sex, Event, Year and remove the inner-outer connections between NOC, Host, Athelete.

A point of particular significance that warrants attention is that NOC and Name exist both in the outside of athelete and inside of athelete. suppose an athlete, A, participates in the Olympic Games. The year in which the athlete participates is denoted as T_1, his or her home country is denoted as C_1, and the time attribute of the Olympics is denoted as T. The country attribute is denoted as C. Then, there is a relation:

$$A \cap T = T_1, A \cap C = C_1$$

This means that NOC and year exist as properties inside Athelete in a certain way, but also have more features which can not be included in Athelete. So they should also appear at the outside of Athelete.

As a result, we can simplify Figure 6 into Figure 7 to make our model clearer.

3.3 Word2vec

In order to quantify variables, we employed word2vec, based on the Continuous Bag of Words (CBOW) and Skip-gram algorithms.

Prior to the administration of training, the context is established by building our exclusive corpus (see Figure 8), which can characterize the degree of similarities and differences between the individual words. In this manner, the textual content provided in the problem can be quantified without any loss of information.

Host:
Paris, a *romantic* city in *Northern France*, is a historic city renowned for its iconic *Eiffel Tower*, world-class art museums like the *Louvre*, charming cafes, and elegant fashion boutiques.

NOC:
Australia's capital is *Canberra*, located at latitude *35.2809°S*, longitude *149.1300°E*. It shares borders with no country but is surrounded by the *Pacific Ocean*. The area of Australia is *7.7 million* square kilometers.

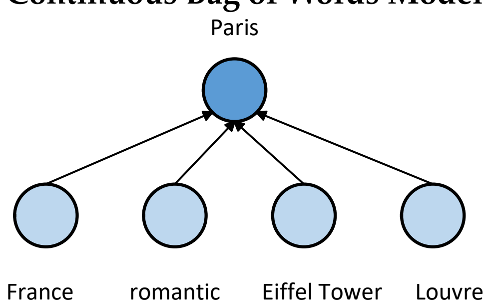
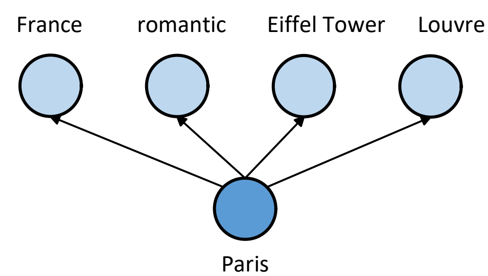
Event:
Archery, marked by the *ARC* code, is a *precision* sport regulated by World Archery, and its competitions are a display of skill and *focus*.

Figure 8: Sentence-making examples(red words are main objectives which should be replaced by vectors, underlined words are information related to red words closely.)

Because the input consists of multiple utterances, so we need to be decomposed into word sequences at first.

Then we can turn words into vectors as follows:

Table 2: Word2vec

Algorithm	**Continuous Bag of Words Model**	**Skip-gram Model**
		
training objective	$\prod_{i\in C} \mathbf{Pr}(\omega_i \vert \omega_j : j \in N+i)$	$\prod_{i\in C} \mathbf{Pr}(\omega_j : j \in N+i \vert \omega_i)$
logarithmic formula	$\sum_{i\in X} \log(\mathbf{Pr}(\omega_i \vert \omega_j : j \in N+i))$	$\sum_{i\in C, j\in N+i} \ln \mathbf{Pr}(\omega_j \vert \omega_i)$
simplifications	$\sum_{i\in C, j\in N+i} \left(v_{\omega_i} \cdot v_{\omega_j} - \ln \sum_{\omega\in V} e^{v_\omega \cdot v_{\omega_j}}\right)$	$\sum_{i\in C, j\in N+i} \left(v_{\omega_i} \cdot v_{\omega_j} - \ln \sum_{\omega\in V} e^{v_\omega \cdot v_{\omega_i}}\right)$

The training objective is thus to maximize the training objective in table2. The objective is to maximize the total probability of the corpus using a model that predicts words based on their neighbors. To address numerical instability in products, we take the logarithm. By simplification, it is necessary to maximize the last line of the equation.

As demonstrated in Figure 9, the quantization results of Host, NOC and Event are finally obtained.

Host:
Paris: (-0.47628674 0.68641096 ... -0.50005627 -0.09373739)
NOC:
Australia: (-0.25766882 0.2604637 ... -0.14131029 0.87454194)
Event:
Archery: (0.17530853 0.9096703 ... 0.10617932 -0.46408582)

100-dimensional vector

Figure 9: Quantitative examples

For Sex, we use two-dimension vector to quantify: $[1, 0]$ for male and $[0, 1]$ for female.

For the four-digit Year, the numerical sequence is segmented into four distinct components: the thousands, hundreds, tens, and digits. These components are regarded as the four dimensions of the four-digit year. To illustrate, the year 2024 can be represented as a four-dimension vector $[2, 0, 2, 4]$.

According to the algorithm specifics delineated in Figure 7, the attributes of the athlete (Name) are determined by Sex, NOC, Event, and Year. Consequently, the properties of Athlete's matrix are comprised of these four elements.

3.4 Random Forest

Random Forests or random decision forests can be divided into two part: construction of decision tree and construction of Random Forest.

Algorithms for constructing decision trees usually work top-down, by choosing a variable at each step that best splits the set of items.[6] Different algorithms use different metrics for measuring "best". These generally measure the homogeneity of the target variable within the subsets. In this paper, we create a new metrice to build decision tree, which makes the tree more balanced by incorporating multiple metrics and introducing denoising factors:

$$\begin{aligned} Score(s) = \alpha \cdot \sqrt{Gain\_ratio(s)(1 - Gini'(s))} \cdot \phi(s|t) \\ + \beta \cdot \frac{CMI(X;Y|Z)}{max(H(X), H(Y))} \cdot RMI(X;Y) \end{aligned} \tag{1}$$

In this metric, $Gain\_ratio$ can be expressed as the equation below:

$$\text{GainRatio}(D,a) = \frac{\text{Gain}(D,a)}{\text{IV}(a)} = -\frac{\text{Ent}(D) + \sum_{v=1}^{V} \frac{|D^v|}{|D|} \cdot \sum_{k=1}^{|\mathcal{Y}|} p_k \log_2 p_k}{\sum_{v=1}^{V} \frac{|D^v|}{|D|} \log_2 \frac{|D^v|}{|D|}} \tag{2}$$

Gini'(s) can be expressed as the equation below:

$$I_G(p) = 1 - \Sigma_{i-1}^{J} p_i^2 \tag{3}$$

denoising factors can be expressed as the equation below:

$$denoising\ factors = \beta \cdot \frac{CMI(X;Y|Z)}{max(H(X),H(Y))} \cdot RMI(X;Y) \tag{4}$$

$$H(X) == \Sigma_{x \in \mathscr{X}} p(x) log_b p(x) \tag{5}$$

and we achieve pruning by using a comprehensive way: The overall approach employs post-pruning, taking into account both the accuracy and the distribution of nodes within the decision tree. Specifically, branches with minor accuracy changes are not pruned if they contain a large number of nodes, whereas branches with significant accuracy changes are pruned if they consist of only a few nodes.

Finally, a comprehensive voting algorithm for Random Forests can be presented:

Through classification accuracy weighting within individual trees, followed by similarity weighting between trees, and further refined by latent nearest neighbor (LNN) weighting, the overall weighting mechanism is systematically enhanced.

The combined weight w_i is calculated as follows:

$$w_i = \sqrt{w_{ij}^{\text{similarity}} \cdot w_i^{\text{LNN}}} = \sqrt{\frac{1}{1 + \sum_{j=1}^{N} d_{edit}(\hat{y}_i, \hat{y}_j) \cdot \frac{1}{1+\alpha \cdot \epsilon_i}} \cdot \frac{1}{1 + d(\hat{y}_i, X_{out})}} \tag{6}$$

Here, each weighting factor further refines the weights which are established in the previous layer.

Using the combined weights, weighted voting is performed to determine the final classification result $\hat{y}$:

$$\hat{y} = \arg\max_j \left(\sum_{i=1}^{N} w_i \cdot \mathbb{I}(\hat{y}_i = j) \right) \tag{7}$$

where $\mathbb{I}(\hat{y}_i = j)$ is an indicator function that equals 1 if tree i predicts class j, and 0 otherwise.

By emplementing these algorithms above into our model, we can express our model by using Simplified pseudocode as below.

Random Forest Regression - Pseudo Code

Input: Data from the years 2016, 2020, and 2024 for training, 2028 data for prediction

1. **For each country folder in the data root path:**
   (a) **For each of the years 2016, 2020, and 2024:**
      i. Load the CSV file
      ii. Extract and pre-process relevant features
      iii. Combine the feature data from years (2016, 2020, 2024) into training dataset $\mathbf{v}_{\text{train}}$
      iv. Extract medal data (gold, silver, bronze, total) and append to target vectors $\mathbf{v}_{\text{gold}}, \mathbf{v}_{\text{silver}}, \mathbf{v}_{\text{bronze}}, \mathbf{v}_{\text{total}}$
      v. input the data into the random forest algorithm and progressively optimize through multiple iterations
   (b) **Else:** If 2028 data does not exist:
      i. Output: "Data for 2028 is missing. Skipping prediction."

Output: 2028 predicted results for each country

3.5 The Result of Model

3.5.1 The Projections for the Medal Table

It was predicted that 93 countries would win medals in the 2028 Olympics. The top 20 countries by Gold's ranking and the top 20 countries by Total's ranking are displayed in Figure 11.

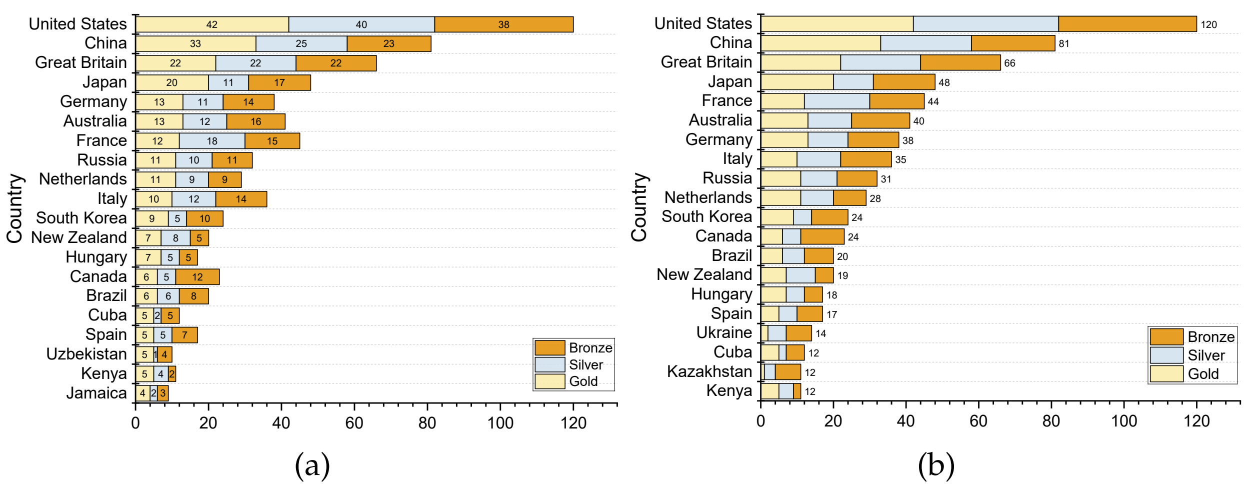


Figure 10: Medal Table: **a)**Predicted 2028 Medal Table ranked by Gold; **b)**Predicted 2028 Medal Table ranked by Total

Among the 84 countries that won medals in 2024, our predictions indicate the following: On the medal table ranked by gold medals, 21 countries are expected to improve their rankings, while 29 countries are anticipated to experience a decline. On the medal table ranked by total medals, 28 countries are predicted to see an improvement in rankings, whereas 42 countries are projected to experience a decline. The rankings for the remaining countries are expected to remain unchanged.

The United States is projected to increase its gold medal count by 2, while experiencing a decrease of 6 in its total medal count. The United Kingdom is expected to gain 8 additional gold medals and 1 additional total medal. In contrast, China is anticipated to see a reduction of 7 gold medals and a decline of 10 in its total medal count.

3.5.2 0 to 1 Breakthough

The results of the model are then used to construct a mapping of the 67 countries that did not win medals in previous years and may achieved a breakthrough of 0-1 in 2028. The five countries with the highest probability (most likely to win a medal) and the corresponding probabilities are as follows:

NOC	Probability
Angola	0.7631
Honduras	0.7463
Samoa	0.6119
Guinea	0.4478
Mali	0.4328

Table 3: 0-1 breakthrough

The remaining 62 countries have a lower probability than even 0.4000 of achieving a 0-1 breakthrough, and it is assumed that they are unable to do so.

3.5.3 Relationship between the events and countries

Countries prioritize their dominant sports programs to maximize medal achievements, with the number of medals serving as an indicator of focus. This part posits a positive correlation between medal count and the level of attention given to specific sports. By analyzing the data of medal from leading sports nations, we identify the key events that each country emphasizes during each Olympic cycle. Selected results are presented below, and Analyses for other years can be conducted in a similar manner:

Table 4: 2016-2024 Focused Sport

Year	Country	Most-focused Sport	Sub-focused Sport	Subsub-focused Sport
2016-2024	USA	Swimming	Athletics	Wrestling
2016-2024	China	Diving	Shooting	Swimming
2016-2024	Great Britain	Athletics	Cycling Track	Swimming

1. **Cause Analysis:** The prioritization of Olympic sports by nations is driven by athlete development systems, cultural history, strategic investments, and coaching quality. Effective systems emphasize early talent identification, structured training, and broad recruitment. Historical engagement and cultural traditions provide foundational advantages. Strategic investments, such as resource allocation to medal-rich events, amplify competitiveness. Elite coaching, integrating advanced methods and mentorship, plays a pivotal role in optimizing performance and ensuring sustained success.

2. **Impact of Events Selection on National Winning Medal Outcomes:** National event strategies are shaped by historical dominance, resource allocation, and cultural contexts, with key approaches as follows:

   (a) **Traditional Advantageous Events as the Pillar of Stable Medals:** Sustained focus on historically strong events ensures a reliable medal source. These events leverage cultural resonance, deep training expertise, and robust talent pools, forming a cornerstone of athletic success.

   (b) **Strategic Event Selection as the Key to Enhancing Comprehensive Competitiveness:** Targeting scalable, culturally significant, or emerging sports enhances medal counts, strengthens global athletic standing, and fosters sustainable development in competitive sports.

   (c) **Diversified Event Portfolio as an Effective Way to Strengthen Overall Strength:** Beyond traditional strengths, expanding into new disciplines increases medal potential, boosts comprehensive competitiveness, and highlights a nation's adaptability across diverse fields.

4 Model Evaluation

Subsequent to the construction of the aforementioned model, an analysis was conducted to assess its accuracy and the extent of its performance.

4.1 Model Precision

1. Precision:

$$Prec = [1 - \frac{1}{n}\sum_{i=1}^{n} |\frac{y_i - \hat{y}_i}{y_i}|] \times 100\% \tag{8}$$

2. the coefficient of determination:

$$R^2 = 1 - \frac{\sum\limits_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum\limits_{i=1}^{n}(y_i - \overline{y})^2} \tag{9}$$

3. The mean square error(MSE) and root mean square error(RMSE):

$$MSE = RMSE^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{10}$$

4. Variation/Emergence Rate:

- **Accuracy:**

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{11}$$

- **Precision:**

$$Precision = \frac{TP}{TP + FP} \tag{12}$$

- **Recall:**

$$Recall = \frac{TP}{TP + FN} \tag{13}$$

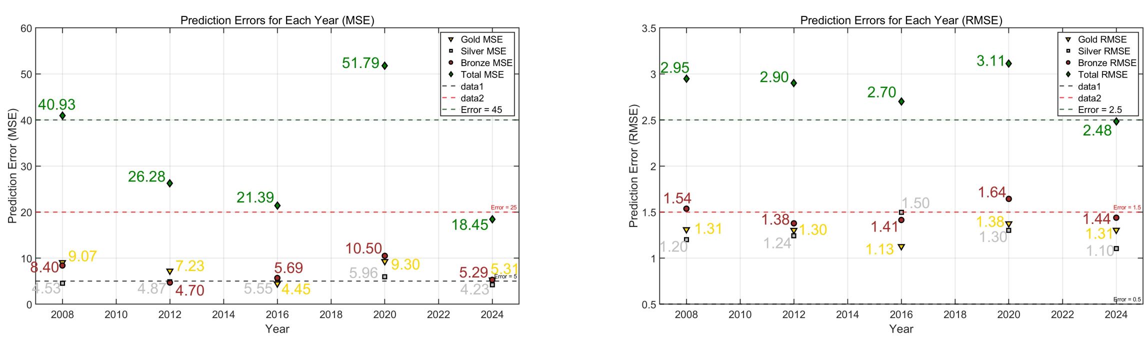
- **F_1 Score:**

$$F_1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \tag{14}$$

Table 5: Confusion Matrix

	Countries actually won medals	**Countries didn't actually win medals**
The country is predicted to have medals.	TP	FP
The country isn't predicted to have medals.	FN	TN

The results of the MSE and RMSE calculations are displayed below:



(a) (b)

Figure 11: Medal Table: **a)**Prediction Errors for Each Year (MSE); **b)**Prediction Errors for Each Year (RMSE)

The model's five-year average Accuracy was calculated to be 89.21%, its Precision 89.56%, its Recall 84.62%, and its F_1 score 86.98%, based on five years of data. These findings indicate that the model demonstrates a high degree of proficiency in accurately predicting medal counts in the majority of cases.

4.2 Level of Performance

1. Cook's Distance

$$D_i = \frac{(e_i^2)}{p \cdot MSE} \cdot \frac{x_i^2}{1 - h_i} \tag{15}$$

Let e_i represent the residual of the ith observation, p the number of model parameters, MSE the mean square error, and h_i the ith row value of the hat matrix. Cook's Distance quantifies the influence of a data point on the regression model's fit. A data point with significantly high Cook's Distance indicates potential outlier behavior, likely impacting model performance.

2. Bayesian information criterion (BIC)

$$BIC = \ln(n)k - 2\ln(L) \tag{16}$$

Where k denotes the number of parameters in the model, L signifies the maximum likelihood estimate, and n represents the number of samples.

3. Adjustment of the coefficient of determination (Adjusted R^2)

$$\text{Adjusted} R^2 = 1 - (1 - R^2) \cdot \frac{n - 1}{n - p - 1} \tag{17}$$

the term n is used to denote the number of samples, while p is used to denote the number of features. The adjusted R^2 is a measure that accounts for the number of variables in the model. The magnitude of the adjusted R^2 is directly proportional to the efficacy of the model fit, with larger values indicating a more robust model fit and a more substantial consideration of the number of variables.

By employing these aforementioned equations above into our model, we illustrate the extent of the model's superior performance.

5 Great Coach Effect

The model predicted medal counts for the 2028 LA Olympics but was found biased due to unquantifiable factors. To enhance the model's precision, we will integrate the Great Coach Phenomenon.

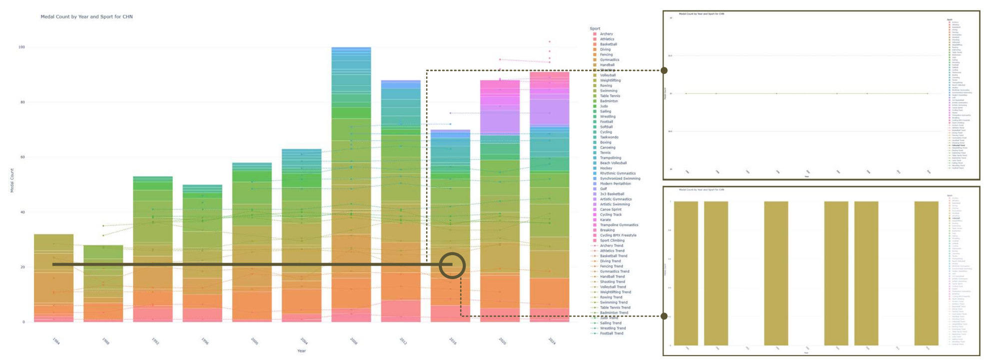
5.1 Data Analysis

Initially, a group of exemplary coaches is selected through a comprehensive review of information regarding exceptional coaches from previous years:

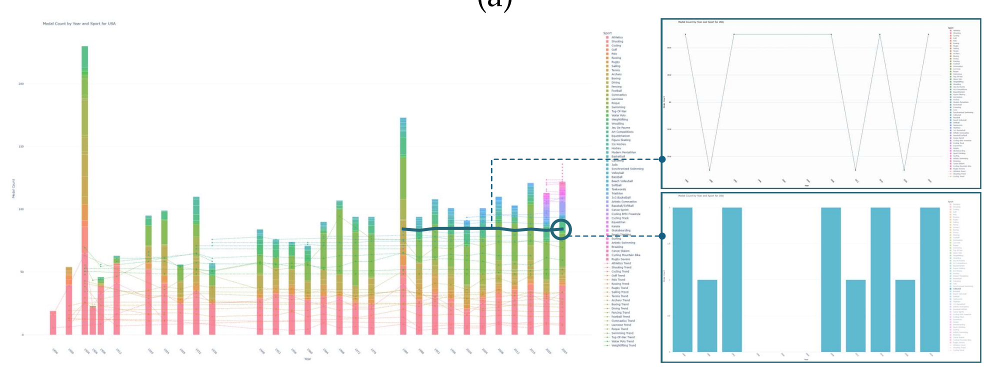
Table 6: Representative "great coaches"

Names of great coaches	Countries in which he/she is a coach	Coaching hours
Lang Ping	China, US	China(1995-1999,2013-2021),US(2005-2008)
Christian Bauer	France, China	France(1990s-2000s), China(2006-2008)
Jeremy Gan	Malaysia, Japan	Malaysia(2010s), Japan(2020s)

Medal trends were analyzed before and after the tenure of renowned coaches. For example, Lang Ping coached China's women's volleyball team from 1995 to 1999, leading to significant medal successes, while the team failed to medal in 1992 and 2000. Similarly, as head coach of the U.S. team from 2005 to 2008, Lang Ping achieved a silver medal and two total medals, compared to no medals in 2004 and one in 2008. From 2013 to 2021, China's women's volleyball team secured a medal in 2016, but when Lang Ping was not in charge of the Chinese team, no medals were earned.



(a)



(b)

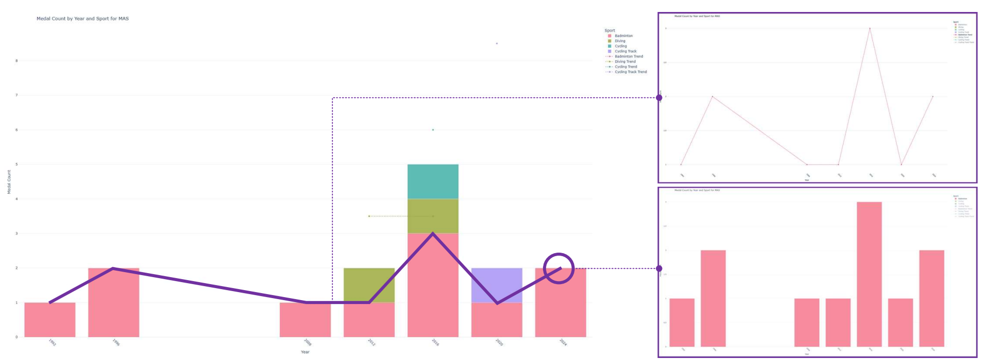
Figure 12: Medal Count by Year and Sport (volleyball) for the country: **a)**CHN; **b)**USA

Using the aforementioned analytical approach, we can infer that the emergence of a "great coach" in a given country is typically associated with the following data characteristics:

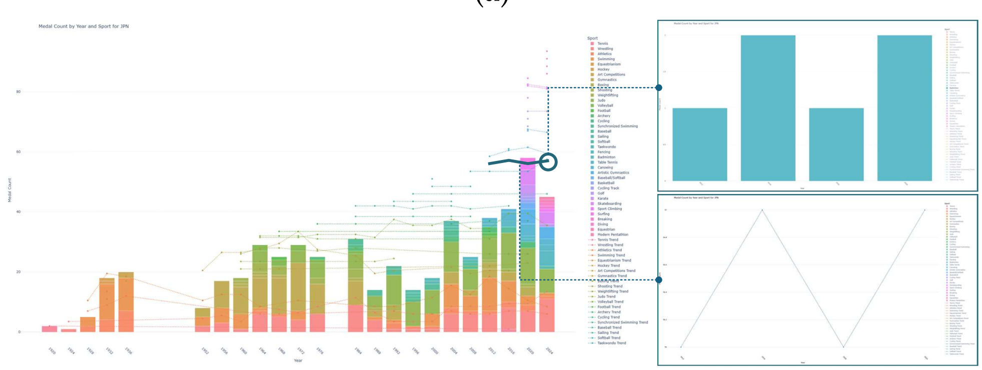
1. The number of medals won by the country during the period when it was coached by the “Great Coach” has stabilized at a value higher than that of other periods.
2. The number of medals won by the country during the period in which it was coached by the "Great Coach" has stabilized at a value that exceeds the number of medals won during other periods.
3. The nation’s medal count under the direction of the renowned coach, known as "Great Coach" has undergone a notable shift from a period of stagnant growth to one of positive progression with each successive Olympic Games.

5.1.1 Data Characteristics

A thorough examination of the medal statistics in badminton has substantiated the existence of the "great coach" phenomenon. For instance, Malaysian badminton witnessed a substantial surge in medals from 2012 to 2016, with 2016 marking the highest number of medals won to date. And Japan has demonstrated a consistent and steady increase in medal acquisitions since 2012.



(a)



(b)

Figure 13: Medal Count by Year and Sport (badminton) for the country: **a)**MAS; **b)**JPA

The hypothesis posited that Malaysia may have "great coaches" from 2012-2016 and Japan from 2020 to the present. A comparative analysis of the data in Table6 reveals that Jeremy Gan has served as the coach for both countries since 2010, thereby validating this prediction and confirming the impact of the "great coach" effect.

5.2 The impact of a "great coach" on the total medal count

The impact of "great coaches" on the total medal count will be assessed from three perspectives:

1. first, trends in the medal count by country
2. second, the impact of other factors on the medal count
3. third, the specific role of "great coaches" on the medal count

5.2.1 Description of trends

The data demonstrate that trends in medal totals vary over time as countries change in strength, technology, and athlete turnover. To illustrate this point, the following list presents United States that achieved the highest number of gold medals in 2024, with the total number of medals awarded over time indicated for each nation.

The modification in the quantity of medals accumulated by the United States can be broadly categorized into three distinct phases:

Phase I: 1896-1912, concluding on the eve of WWI. The U.S. medal count experienced significant fluctuations, peaking in 1904, a record yet to be surpassed. This period reflects the era's complex and evolving global dynamics.

Phase II: 1920-1936, commencing with the cessation of World War I and concluding with the onset of World War II. Medal growth was steady, with notable increases in 1920, 1924, and 1932, culminating in a substantial rise in 1936 compared to 1928.

Phase III: 1948-present, commencing with the cessation of World War II. The United States' medal count demonstrate a period of substantial growth, which have largely maintained a high level.

Since its Olympic debut, the United States has expanded its participation across events and steadily increased its medal count. This trajectory mirrors global developments in competitive sports, affirming the nation's pivotal role on the international stage.

5.2.2 Extent of impact

As previously discussed, the model incorporates all relevant factors, including country, time, host, and athlete. The model accounts for nearly all effects on medal totals with the exception of the influence of "great coaches."

The model fails to account for the "great coach" effect, resulting in discrepancies between predicted and actual medals. This effect can be quantified through normalization.Let the predicted medals for a given year be $\hat{y_{ij}}$ and the actual medals be y_{ij}, then the impact can be expressed as follow:

$$impact = \frac{1}{n}\Sigma_{j=1}^{n}|\frac{y_{ij} - \hat{y_{ij}}}{y_{ij} + \epsilon}| \cdot 100\% + S(y_{ij}, \hat{y_{ij}}, r, \epsilon', \lambda, k, \sigma_y, \beta_1, ..., \beta_h) \tag{18}$$

$$S(y_{ij}, \hat{y_{ij}}, r, \epsilon', \lambda, k, \sigma_y, \beta_1, ..., \beta_h) = \lambda \cdot (1 - e^{-k|y_{ij} - \hat{y_{ij}}|}) + r \cdot \Sigma_{f=2}^{h}\beta_f \cdot [\frac{y_{ij} - \hat{y_{ij}}}{(\sigma_y + \epsilon') \cdot y_{ij}}]^h \tag{19}$$

- n: Total number of data points used to calculate the average of the overall error.
- ϵ: Small constants that prevent the denominator from being zero are used to improve the numerical stability of the formula.
- ϵ': Used as a stability parameter in the denominator of higher order terms to avoid instability in the formula caused by too small values.
- σ_y: The standard deviation of the dataset, which is used to characterize the range of the distribution of the true value y_{ij}, helps to adjust the dynamics of the higher-order noise correction.

The physical meaning of the formula

1. **The main part of the formula**: Measure the degree of deviation between the predicted value $\hat{y_{ij}}$ and the true value y_{ij} and use averaging to minimize the effect of single-point outliers.
2. **Index modifiers** ($\lambda \cdot (1 - e^{-k|y_{ij} - \hat{y_{ij}}|})$): Smoothing errors and reducing the drastic effect of noise on overall deviation.
3. **Higher-order modifiers** ($r \cdot \sum_{f=2}^{h} \beta_f \cdot [\ldots]^h$): Fine-grained corrections for more complex noise distributions improve the robustness of the formula to outliers.

5.3 Great Coach Investment

The following is a summary of the criteria for evaluating the investment decision regarding the Great Coach Investment:

1. Sport's Programs with a positive trend in total medals but a relatively slow rate of growth
2. Sport's Programs that show volatility in medal scores
3. Sport's Programs that have experienced a peak in medals but have seen a recent decline in performance
4. Programs that have maintained a steady state of medals over time

In accordance with the aforementioned guidelines, we choose three countries to identify sports where they should consider investing in a "great" coach and estimate that impact:

1. Brazil
   (a) choice: prioritize the investment in high-caliber coaches within the following disciplines: Athletics, Swimming, Football, Judo, and Beach Volleyball.
   (b) reason: Brazil is experiencing a period of growth and development in Athletics and Judo, though with fluctuations in performance. Swimming did not achieve a medal victory in 2024, and Football has remained relatively stable, having reached its zenith. Conversely, Beach Volleyball has experienced a significant decline in its competitive standing since 2016.
2. Norway
   (a) choice: focus should center on Athletics, Shooting, Sailing, and Rowing.
   (b) reason: While Athletics has exhibited a decline in performance, Shooting and Sailing have experienced a notable decline since 1920, and Rowing has maintained a steady trajectory.
3. Indonesia
   (a) choice: prioritize investment in badminton and weightlifting.
   (b) reason: Indonesia's badminton shows a downward trend, and weightlifting has a big volatility.

The following scheme has been developed to evaluate the investment results corresponding to the aforementioned criteria. This scheme is based on an in-depth analysis of the correlation and impact of "great coaches" and data characteristics.

Criteria 1:

$$medal' = medal_{last\ year} + \epsilon \cdot growth\_rate$$

$$growth\_rate = \frac{1}{n}\Sigma_{i=1}^{n} growth\_rate_i (growth\_rate_1 < \cdots < growth\_rate_n)$$

Criteria 2:

$$medal' = medal_{last\ year} + \epsilon \cdot growth\_rate$$

$$growth\_rate = \frac{1}{n}\Sigma_{j=1}^{m} growth\_rate_j (0 < growth\_rate_i)$$

Criteria 3:

$$medal' = \frac{medal_{last\ year} + medal_{max}}{2}$$

Criteria 4:

$$medal' = \{x | x \in \{medal_{stable\ value} + 1, ..., medal_{stable\ value} + top\}\},$$

top to be determined on a case-by-case basis

Substituting the values for the three countries under consideration yields the following predicted investment results:

Table 7: result of three countries' Great Coach Investment

Country	Investment Project	Outcome
Brazil	Athletics	3 medals won, a relative increase of 1 medal
Brazil	Swimming	3 medals won, a relative increase of 1 medal
Brazil	Football	2 medals won, a relative increase of 1 medal
Brazil	Judo	5 medals won, a relative increase of 1 medal
Brazil	Beach Vollyball	2 medals won, a relative increase of 1 medal
Norway	Athletics	4 medals won, a relative increase of 1 medal
Norway	Shooting	4 medals won, a relative increase of 3 medal($\epsilon = \frac{1}{2}$)
Norway	Sailing	2 medals won, a relative increase of 1 medal($\epsilon = \frac{1}{2}$)
Norway	Rowing	2 medals won, a relative increase of 1 medal
Indonesia	Badminton	2 medals won, a relative increase of 1 medal($\epsilon = \frac{1}{2}$)
Indonesia	Weightlifting	32medals won, a relative increase of 1 medal($\epsilon = \frac{1}{2}$)

6 Original Insights

6.1 Revealable factors

In addition to the athlete's personal data, numerous external factors have the capacity to influence sports performance. The utilization of modeling facilitates the revelation of intricate factors, such as the impact of senior athletes on younger generations. This contributes to a more comprehensive comprehension of the dynamics underlying sports performance.

1. **The Impact of Sudden Events:** Major historical events, such as world wars, the COVID-19 pandemic, or political changes (e.g., national dissolution or mergers), have disrupted sports events, leading to the postponement or cancellation of competitions like the Olympics. These disruptions not only affect immediate preparations but also influence the long-term development of sports.

2. **Interaction Between Events in the Same Discipline:** Different events within the same discipline, like short- and long-distance running, are often interdependent. Success or failure in one event can drive or hinder the development of others. Shared training principles can lead to breakthroughs in one event that positively impact others, while negative events may have a cascading effect across the discipline. Analytical models offer insights into these interactions.

3. **Host Venues and the "Host Advantage" Effect:** Host nations gain psychological and environmental advantages, such as cultural familiarity and strong home support, resulting in the "host advantage." This can boost local athletes' performance

but may raise concerns about fairness. Over-reliance on these advantages may hinder athletes' ability to perform in foreign settings.

4. **Monopoly in Specific Events by Certain Nations:** Some countries dominate certain sports, creating competitive barriers for others. For example, China's dominance in table tennis challenges international athletes both technically and psychologically. Analytical models can uncover the mechanisms of such dominance and suggest strategies for promoting diversity and equity in global competitions.

6.2 Sensible Training and Development Strategies to NOC

The findings of these analyses will facilitate the enhancement of athletes' individual performance and provide a scientific foundation for the NOC to develop more effective sports development strategies.

1. **Establishing an Information Processing Center:** The NOC should integrate sports and societal data to monitor global changes, focusing on external factors affecting athletes. A data platform ensures timely strategy adjustments for unpredictable challenges.
2. **Cross-Event and Cross-Disciplinary Collaboration:** Exchanges across events and disciplines promote sharing experiences and innovations, enhancing performance and coordination through collaborative training and cooperation.
3. **Optimizing Competitive Atmosphere:** Home events can increase pressure on athletes and audiences. Regulating the venue atmosphere and providing psychological support helps maintain mental stability and optimal performance.
4. **Fostering Shared Progress in Sports:** Influential nations should assist others by sharing expertise and collaborating to reduce imbalances, as exemplified by China's promotion of table tennis globally through training programs and knowledge exchange.

7 Strengths and Weaknesses

7.1 Strengths

1. **High Predictive Accuracy:** The proposed Word2Vec–Random Forest framework achieves strong predictive performance in Olympic medal forecasting, demonstrating high reliability across historical Olympic datasets.
2. **Generalizability:** Our model can be applied to different Olympic years and various national medal prediction tasks, showing broad applicability and adaptability.
3. **Robustness:** Sensitivity analysis and historical validation confirm the robustness and stability of the model under varying data conditions.

7.2 Weaknesses

1. **Data Dependency:** The accuracy of the model depends heavily on the quality and completeness of historical Olympic data.

2. **Limited External Factors:** Certain factors such as political events, injuries, and psychological conditions are difficult to quantify and are not fully considered in the model.

8 Conclusions and Future Work

This paper proposes an Olympic medal prediction framework based on Word2Vec and Random Forest regression. By combining athlete information, event characteristics, host effects, and historical medal data, the model effectively predicts medal distributions for the 2028 Olympic Games.

The results demonstrate strong predictive accuracy and robustness. In addition, the analysis of host effects, sport specialization, and the "Great Coach" effect provides useful insights for National Olympic Committees (NOCs) in optimizing training strategies and sports investment.

Future work will focus on incorporating more socioeconomic factors and advanced machine learning methods to further improve prediction accuracy and model interpretability.

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